

A Additional Experiments and Analysis

A.1 Training details

textbf{Front reconstruction loss.} The front reconstruction module is designed to recover reliable features from corrupted inputs while preserving their underlying semantic structure. Instead of only using a simple reconstruction objective, we explicitly introduce multiple complementary constraints to stabilize feature recovery from different perspectives.

Let $\mathbf{x}_t^k$ denote the target clean feature of modality k , $\hat{\mathbf{x}}_t^k$ denote the final reconstructed feature, and $\mathbf{x}_{\text{im},t}^k$ denote the intermediate reconstructed feature. The overall loss of the front module is defined as

$$\mathcal{L}_{\text{main}}^k = 0.20 \mathcal{L}_{\text{mse},1}^k + 0.20 \mathcal{L}_{\text{mse},2}^k + 0.15 \mathcal{L}_{\text{cos}}^k + 0.10 \mathcal{L}_{\text{sep}}^k + 0.15 \mathcal{L}_{\text{bce}}^k + 0.20 \mathcal{L}_{\text{cov}}^k. \quad (1)$$

where each term is defined as follows.

Value-level reconstruction losses. To ensure numerical consistency during recovery, we employ two mean squared error losses:

$$\mathcal{L}_{\text{mse},1}^k = \left\| \hat{\mathbf{x}}_t^k - \mathbf{x}_t^k \right\|_2^2, \quad (2)$$

$$\mathcal{L}_{\text{mse},2}^k = \left\| \mathbf{x}_{\text{im},t}^k - \mathbf{x}_t^k \right\|_2^2. \quad (3)$$

The first term constrains the final reconstructed feature to approach the clean target, while the second term regularizes the intermediate reconstruction process and improves optimization stability.

Cosine consistency loss. To preserve the directional structure of the feature in the embedding space, we further introduce a cosine similarity constraint:

$$\mathcal{L}_{\text{cos}}^k = 1 - \frac{\langle \hat{\mathbf{x}}_t^k, \mathbf{x}_t^k \rangle}{\left\| \hat{\mathbf{x}}_t^k \right\|_2 \left\| \mathbf{x}_t^k \right\|_2}. \quad (4)$$

This term prevents the reconstructed feature from deviating excessively in direction even when the Euclidean reconstruction error is small.

Separation loss. To enhance the discriminability of the reconstructed representation, we further impose a separation constraint between positive and negative samples:

$$\mathcal{L}_{\text{sep}}^k = \max \left(0, m + d(\hat{\mathbf{x}}_t^k, \mathbf{x}_t^{k,+}) - d(\hat{\mathbf{x}}_t^k, \mathbf{x}_t^{k,-}) \right), \quad (5)$$

where m is a margin, $\mathbf{x}_t^{k,+}$ denotes the positive target, $\mathbf{x}_t^{k,-}$ denotes the negative target, and $d(\cdot, \cdot)$ is a distance function. This term encourages the recovered feature to remain closer to semantically consistent samples while staying away from inconsistent ones.

Mask prediction loss. Since the front module also predicts unreliable or corrupted positions, we use a binary cross-entropy loss for supervision:

$$\mathcal{L}_{\text{bce}}^k = -\frac{1}{N} \sum_{j=1}^N \left[m_j^k \log \hat{m}_j^k + (1 - m_j^k) \log(1 - \hat{m}_j^k) \right], \quad (6)$$

where $\hat{m}_j^k$ and m_j^k denote the predicted and ground-truth mask value at position j , respectively. This term helps the model explicitly identify unreliable entries and improves robustness under missingness and sparse corruption.

Covariance regularization. To avoid degenerate reconstruction patterns and preserve the overall feature distribution, we additionally regularize the covariance structure:

$$\mathcal{L}_{\text{cov}}^k = \left\| \text{Cov}(\hat{\mathbf{X}}^k) - \text{Cov}(\mathbf{X}^k) \right\|_F^2, \quad (7)$$

where $\hat{\mathbf{X}}^k$ and $\mathbf{X}^k$ denote the reconstructed and target feature sets in a mini-batch, respectively, and $\| \cdot \|_F$ is the Frobenius norm. This term encourages the recovered features to maintain a distributional structure similar to that of the clean features, thereby reducing representation collapse or over-smoothed solutions.

Overall, these losses jointly constrain the front reconstruction module from the perspectives of value recovery, directional consistency, discriminative separation, corruption-position prediction, and distribution preservation. Therefore, the front module does not merely denoise or impute corrupted entries, but also preserves the semantic geometry of the original representation and reduces feature drifting during reconstruction.

Subsequent semantic refinement loss. After obtaining the reconstructed features, the subsequent module further performs semantic alignment and discriminative refinement. Its optimization objective consists of triplet loss, composite loss, and regularization loss:

$$\mathcal{L} = \mathcal{L}_t + \mathcal{L}_c + \mathcal{L}_r. \quad (8)$$

The triplet loss explicitly enlarges the inter-class margin and improves feature discrimination across modalities:

$$\mathcal{L}_t = t(\theta_{a_i}^+, \theta_{v_i}^+, o_{a_i}^-) + t(\theta_{a_i}^+, \theta_{v_i}^+, o_{v_i}^-) + t(\theta_{w_i}^-, \theta_{a_i}^+, o_{w_i}) + t(\theta_{w_i}^-, \theta_{v_i}^+, o_{w_i}). \quad (9)$$

where $t(\cdot)$ denotes the triplet loss function. This term encourages samples from different classes to be more separable in the shared space, which is especially important for unseen-class recognition in the GZSL setting.

The composite loss combines reconstruction consistency and cross-modal alignment:

$$l_{\text{rec}} = d(\rho_{a_i}^i, w_i) + d(\rho_{v_i}^i, w_i) + d(\rho_{w_i}^i, w_i), \quad (10)$$

$$l_{\text{ct}} = t(\rho_{w,i}^+, \rho_{a,i}^+, \rho_{a,i}^-) + t(\rho_{w,i}^+, \rho_{v,i}^+, \rho_{v,i}^-), \quad (11)$$

$$l_w = t(\theta_{w,i}^+, \theta_{a,i}^+, \theta_{a,i}^-) + t(\theta_{w,i}^+, \theta_{v,i}^+, \theta_{v,i}^-) + t(\theta_{a,i}^+, \theta_{w,i}^+, \theta_{w,i}^-) + t(\theta_{v,i}^+, \theta_{w,i}^+, \theta_{w,i}^-), \quad (12)$$

and the composite loss is written as

$$\mathcal{L}_c = l_{\text{rec}} + l_{\text{ct}} + l_w, \quad (13)$$

where, l_{rec} enforces decodability by minimizing the distance between decoded features and semantic embeddings, while l_{ct} and l_w further align different modalities through triplet-style cross-modal constraints. In this way, the model learns not only discriminative representations but also a more consistent audio-visual-semantic shared space.

Table 1: Hyper-parameter settings used on the three datasets.

	VGG-GZSL	UCF-GZSL	ActivityNet-GZSL
Batch Size	256	256	256
Epochs	50	50	50
Initial Learning Rate	1e-3	1e-3	1e-3
Learning Rate Scheduler	✓	✓	✓
Latent Dimension	64	16	64
Hidden Dimension	512	512	512
Decoder Dropout	0.2	0.2	0.2
Embedding Dropout	0.2	0.2	0.2
Additional Dropout	0.1	0.1	0.1
Transformer Depth	1	1	1
Number of Frequencies	16	8	8
Triplet Margin	1.0	1.0	1.0
Momentum	0.99	0.99	0.99
Number of Batches	500	500	500
Seed	42	42	42

The regularization loss is used to preserve semantic anchoring while reducing unnecessary deviation during feature transformation:

$$\mathcal{L}_r = d(\phi_{a,i}^{\text{rec}}, \phi_{a,i}) + d(\phi_{v,i}^{\text{rec}}, \phi_{v,i}) + d(\phi_a^i, \phi_w^i) + d(\phi_v^i, \phi_w^i). \quad (14)$$

This term promotes the alignment between reconstructed embeddings, modality-specific embeddings, and text embeddings, thereby preserving modality-specific information while improving semantic transferability.

Stage-wise training strategy. The whole model is trained in three stages. In the first stage, we optimize only the front reconstruction module while freezing the subsequent semantic refinement module, so that the model first learns to restore corrupted inputs and stabilize feature quality. In the second stage, we freeze the front module and optimize only the subsequent refinement module, allowing semantic alignment and discriminative objectives to be learned on top of relatively stabilized reconstructed features. In the final stage, the whole network is jointly fine-tuned to further improve the compatibility between reconstruction and semantic refinement.

Moreover, since the front module and the subsequent module serve different purposes and exhibit different optimization behaviors, we use two separate optimizers for them during training. This design improves optimization stability and allows each part to converge according to its own objective characteristics, while the final joint fine-tuning further enhances end-to-end consistency.

A.2 Experimental parameter settings

Batch Size denotes the number of samples used in each optimization step and mainly affects training stability, computational efficiency, and memory usage. Epochs denotes the number of complete passes over the training set and determines the overall training duration and convergence degree. Initial Learning Rate controls the step size at the beginning of optimization and is critical to convergence speed and stability. Learning Rate Scheduler indicates whether the learning rate is dynamically adjusted during training to balance fast optimization and stable convergence. Latent Dimension denotes the

dimensionality of the latent representation and reflects the abstraction capacity of the model. Hidden Dimension denotes the dimensionality of intermediate hidden representations in the network and affects its representation capability. Decoder Dropout, Embedding Dropout, and Additional Dropout are regularization terms applied to different modules to alleviate overfitting and improve robustness. Transformer Depth denotes the number of stacked Transformer layers, while Number of Frequencies denotes the number of frequencies used in positional or frequency encoding. Triplet Margin is the margin parameter in the triplet loss and is used to enhance feature discriminability. Momentum denotes the coefficient used to smooth optimization updates through historical gradients. Number of Batches denotes the number of mini-batches in each epoch, and Seed controls stochastic processes such as initialization and data shuffling for reproducibility.

A.3 Effect of noise intensity on model robustness

Following the standard protocols of generalized zero-shot learning (GZSL), we report four evaluation metrics, namely *Seen*, *Unseen*, *HM*, and *ZSL*, to provide a comprehensive assessment of model performance. Specifically, *Seen* denotes the classification accuracy on test samples from seen classes, which measures the model’s ability to preserve discriminative knowledge learned from categories observed during training. In contrast, *Unseen* denotes the classification accuracy on test samples from unseen classes, reflecting the model’s ability to transfer semantic or cross-modal knowledge from seen categories to novel categories that are not accessible during training. Since a GZSL model is required to perform well on both seen and unseen classes simultaneously, evaluating only one side is insufficient. Therefore, we additionally report the harmonic mean (*HM*) of *Seen* and *Unseen*, defined as

$$HM = \frac{2 \times \text{Seen} \times \text{Unseen}}{\text{Seen} + \text{Unseen}},$$

which serves as a balanced indicator of the trade-off between seen-class recognition and unseen-class generalization. A high HM score indicates that the model does not overly favor either the seen or the unseen categories. In addition, we report *ZSL*, which denotes the conventional zero-shot learning accuracy evaluated on a test set containing only unseen classes. Unlike *Unseen* in the GZSL setting, *ZSL* excludes the competition from seen classes and therefore reflects the model’s pure generalization capability on novel categories under the standard zero-shot setting.

We conduct experiments on three widely used audio-visual generalized zero-shot learning benchmarks, namely *UCF-GZSL*, *VGGSound-GZSL*, and *ActivityNet-GZSL*. These datasets provide paired visual and audio information together with class-level semantic descriptions, making them suitable for evaluating cross-modal transfer and zero-shot generalization.

UCF-GZSL is built upon the UCF101 action benchmark and is widely used in audio-visual zero-shot learning. It contains human action categories with relatively clear semantic boundaries and moderate complexity, making it suitable for evaluating basic cross-modal alignment and knowledge transfer ability.

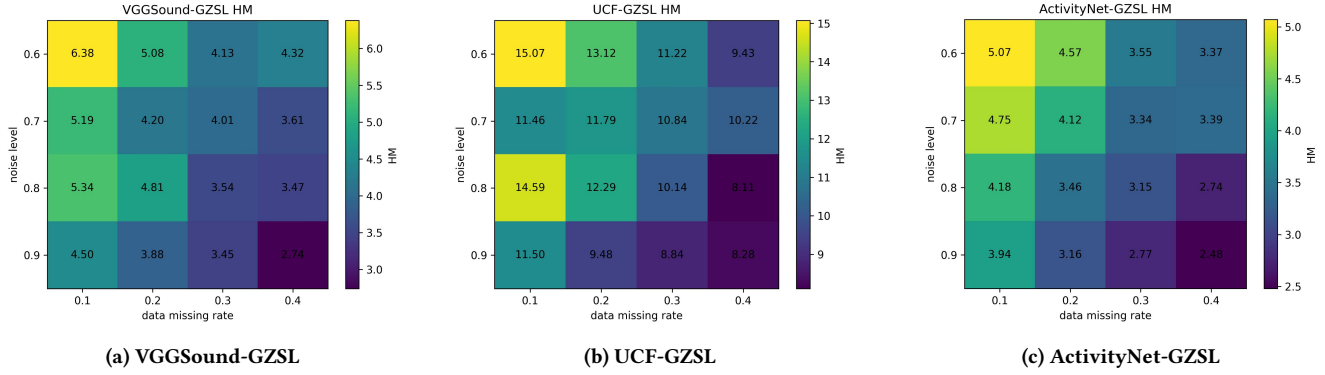


Figure 1: Heatmaps of HM under different data missing rates and noise levels on three datasets.

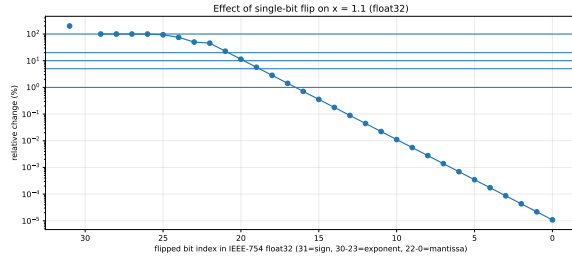


Figure 2: IEEE-754 effect of single-bit flip

VGGSound-GZSL is derived from VGGSound and provides a more challenging and realistic testbed. It covers diverse sound-producing events in unconstrained environments, with richer acoustic variation, more complex backgrounds, and stronger inter-class ambiguity, thus placing higher demands on representation robustness and modality interaction.

ActivityNet-GZSL is constructed from ActivityNet and focuses on more complex activity understanding scenarios. Compared with UCF-GZSL, it involves longer temporal duration, larger scene variation, and more complicated event composition, which makes cross-modal evidence extraction and unseen-class generalization more challenging.

A.4 Noise robustness analysis

As shown in the Fig. 1, HM decreases progressively as the corruption level increases, and the degradation caused by additive noise is generally more pronounced than that caused by missingness. This suggests that continuous perturbation on feature values tends to affect the overall representation structure more directly, whereas missingness mainly removes partial information and does not always disrupt the global feature geometry to the same extent. To further understand the practical effect of discrete bit-level corruption, Fig. 2 additionally presents a float32 case study, where each individual bit of an example value is flipped and the resulting numerical variation is measured. The resulting plot shows that single-bit perturbations usually fall into two categories: they either cause only very small value changes and thus have limited influence on the underlying feature structure, or produce relatively obvious

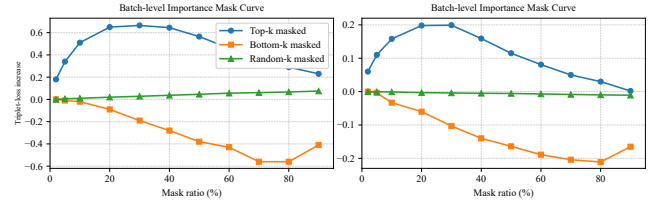


Figure 3: Explanation of missing dimensions

abnormal deviations that are easier to identify as unreliable positions. This observation is consistent with our empirical finding that mild bit-level perturbations do not lead to noticeable fluctuations in downstream performance. Based on this, we do not treat low-probability single-bit corruption as a sufficiently effective source of task-relevant structural degradation. Instead, we adopt a stronger bit-flip simulation strategy by inverting the full binary pattern, so that the induced corruption can produce a more meaningful disturbance to the downstream task and provide a clearer evaluation of model robustness under severe discrete distortion. As shown in Fig. 3, masking the top-ranked dimensions selected according to gradient magnitude leads to a much larger loss increase than masking the bottom-ranked or randomly selected dimensions, especially in the low-to-moderate mask-ratio regime. This indicates that the dimensions with larger gradient responses indeed carry more task-relevant and discriminative information. However, when the mask ratio becomes further larger, the gap gradually shrinks and the top- k curve starts to fall back. We attribute this behavior not to the disappearance of feature importance, but to the fact that the representation has already lost a substantial amount of discriminative freedom under heavy masking. In such a high-dimensional sparse regime, the model is no longer sensitive to which additional dimensions are removed, and the corresponding loss signal becomes less informative. Based on this observation, overly small missing rates are not sufficiently challenging, while overly large missing rates tend to drive the task into a saturated degradation regime. Therefore, a moderate missing-rate range is more appropriate, since it can simultaneously preserve task difficulty and maintain meaningful sensitivity to feature importance.

Table 2: Comparison under different parameter settings.

Method	VGGSound-GZSL				UCF-GZSL				ActivityNet-GZSL			
	S	U	HM	ZSL	S	U	HM	ZSL	S	U	HM	ZSL
Missing rate = 0.1, AWGN intensity = 0.6, Bit-flip rate = 0.1												
AVCA	1.32	0.96	1.11	2.09	4.17	7.76	5.42	12.68	2.42	0.88	1.29	3.43
MSTR	1.31	1.07	1.18	2.23	7.89	6.19	6.93	12.60	4.24	1.25	1.93	4.06
DAAN	2.95	1.74	2.19	2.55	9.38	8.58	8.96	14.86	3.47	2.51	2.92	3.38
QARM	9.68	4.75	6.38	5.65	17.74	13.10	15.07	19.18	6.58	4.13	5.07	4.28
Missing rate = 0.1, AWGN intensity = 0.7, Bit-flip rate = 0.1												
AVCA	1.81	0.72	1.03	2.03	7.04	8.00	7.49	12.78	4.75	3.40	3.96	3.64
MSTR	1.56	0.96	1.19	2.41	9.67	10.35	10.00	16.25	2.54	2.98	2.74	3.91
DAAN	1.97	1.71	1.83	2.41	6.63	8.91	7.61	12.35	2.75	3.02	2.88	4.28
QARM	5.95	4.43	5.08	4.96	15.36	11.45	13.12	17.48	4.60	4.55	4.57	4.81
Missing rate = 0.1, AWGN intensity = 0.8, Bit-flip rate = 0.1												
AVCA	3.45	1.54	2.13	3.01	5.41	5.50	5.45	12.14	2.26	2.01	2.13	2.98
MSTR	0.97	1.13	1.04	2.58	12.76	6.97	9.02	15.73	2.82	2.44	2.62	3.63
DAAN	1.76	2.14	1.94	2.43	6.68	8.29	7.40	11.87	2.78	2.55	2.66	3.60
QARM	5.04	3.51	4.13	4.20	15.72	8.72	11.22	15.30	3.80	3.33	3.55	3.48
Missing rate = 0.1, AWGN intensity = 0.9, Bit-flip rate = 0.1												
AVCA	1.27	1.95	1.54	2.95	4.60	4.25	4.43	11.82	1.01	1.31	1.21	2.65
MSTR	1.27	1.71	1.46	2.70	3.89	6.96	4.99	12.40	1.06	1.27	1.16	2.20
DAAN	1.17	1.28	1.22	2.67	4.31	3.21	3.68	11.95	0.94	1.77	1.23	2.76
QARM	6.15	3.33	4.32	3.80	11.79	7.86	9.43	11.41	3.79	3.03	3.37	3.14
Missing rate = 0.2, AWGN intensity = 0.6, Bit-flip rate = 0.1												
AVCA	2.11	1.16	1.50	3.01	7.38	4.53	5.62	10.51	1.57	1.19	1.35	3.05
MSTR	3.10	1.54	2.06	3.29	12.33	7.08	8.99	16.18	1.06	1.27	1.16	2.20
DAAN	1.76	2.14	1.94	2.43	6.68	8.29	7.40	11.87	2.78	2.55	2.66	3.60
QARM	6.95	4.14	5.19	4.41	15.95	8.95	11.46	14.99	5.47	4.20	4.75	4.52
Missing rate = 0.2, AWGN intensity = 0.7, Bit-flip rate = 0.1												
AVCA	1.45	0.90	1.11	2.00	6.36	3.68	4.67	11.21	1.47	1.01	1.19	2.28
MSTR	2.55	0.87	1.30	2.49	13.63	4.53	6.80	15.88	2.69	1.60	2.01	2.66
DAAN	1.79	2.03	1.90	2.75	7.91	9.66	8.70	12.85	2.31	2.21	2.26	3.42
QARM	5.24	3.51	4.20	3.57	15.86	9.38	11.79	14.73	4.71	3.66	4.12	3.90
Missing rate = 0.2, AWGN intensity = 0.8, Bit-flip rate = 0.1												
AVCA	2.61	1.45	1.86	2.67	4.91	5.26	5.08	11.48	2.08	0.23	0.41	2.77
MSTR	2.86	1.21	1.70	2.98	6.56	4.60	5.41	12.66	1.13	0.90	1.00	2.61
DAAN	2.00	1.91	1.81	2.46	7.15	8.34	7.69	11.87	2.61	2.35	2.47	3.44
QARM	5.03	3.33	4.01	3.62	11.99	9.89	10.84	13.93	3.32	3.36	3.34	3.46
Missing rate = 0.2, AWGN intensity = 0.9, Bit-flip rate = 0.1												
AVCA	1.19	1.72	1.41	2.57	4.19	6.05	5.29	11.62	2.39	0.11	0.21	2.02
MSTR	1.13	1.65	1.34	2.38	3.85	6.96	4.95	11.98	0.96	1.90	1.27	2.72
DAAN	1.17	1.28	1.22	2.67	4.31	3.21	3.68	11.95	0.94	1.77	1.23	2.76
QARM	4.63	2.96	3.61	3.68	12.10	8.85	10.22	13.05	3.66	3.16	3.39	3.60
Missing rate = 0.3, AWGN intensity = 0.6, Bit-flip rate = 0.1												
AVCA	3.45	1.54	2.13	3.01	5.41	5.50	5.45	12.14	2.26	2.01	2.13	2.98
MSTR	0.97	1.13	1.04	2.58	12.76	6.97	9.02	15.73	2.82	2.44	2.62	3.63
DAAN	1.58	1.51	1.54	2.12	3.99	4.13	4.06	14.46	1.25	1.57	1.39	2.67
QARM	7.15	4.26	5.34	4.58	15.24	14.00	14.59	18.78	4.70	3.76	4.18	3.94

Table 3: Comparison under different parameter settings.

Method	VGGSound-GZSL				UCF-GZSL				ActivityNet-GZSL			
	S	U	HM	ZSL	S	U	HM	ZSL	S	U	HM	ZSL
Missing rate = 0.3, AWGN intensity = 0.7, Bit-flip rate = 0.1												
AVCA	1.45	0.72	0.97	2.26	4.42	5.95	5.07	12.02	2.23	0.96	1.35	2.89
MSTR	0.81	3.22	1.30	3.59	9.93	2.56	4.07	13.11	3.07	1.83	2.29	4.22
DAAN	1.40	1.48	1.44	1.97	4.59	4.67	4.63	12.86	1.11	1.38	1.23	2.59
Missing rate = 0.3, AWGN intensity = 0.8, Bit-flip rate = 0.1												
AVCA	3.09	2.55	2.79	3.25	5.09	6.67	5.78	10.25	1.67	1.27	1.44	2.77
MSTR	2.35	0.72	1.11	2.99	8.28	5.87	6.87	11.78	2.47	1.32	1.72	2.64
DAAN	1.64	1.57	1.60	2.20	6.85	7.08	6.97	10.88	2.75	2.29	2.50	3.29
QARM	6.95	3.68	4.81	4.29	13.35	11.38	12.29	16.40	3.09	3.94	3.46	3.99
Missing rate = 0.3, AWGN intensity = 0.9, Bit-flip rate = 0.1												
AVCA	1.91	2.46	2.15	2.75	5.42	4.03	4.62	9.37	1.39	1.43	1.41	2.71
MSTR	0.91	1.33	1.08	2.12	4.60	4.96	4.77	11.03	1.14	1.02	1.08	2.39
DAAN	1.30	1.01	1.14	1.86	3.55	3.26	3.40	12.30	0.88	1.23	1.03	2.74
QARM	3.74	3.36	3.54	3.59	12.28	8.63	10.14	15.80	3.19	3.12	3.15	3.47
Missing rate = 0.4, AWGN intensity = 0.6, Bit-flip rate = 0.1												
MSTR	1.45	1.48	1.46	2.61	5.40	6.46	5.88	11.59	1.25	1.53	1.38	2.42
DAAN	1.24	1.25	1.24	1.88	3.85	6.30	4.78	13.20	1.30	1.43	1.36	3.01
QARM	7.47	3.22	4.50	4.03	12.50	10.65	11.50	13.38	4.27	3.66	3.94	3.97
Missing rate = 0.4, AWGN intensity = 0.7, Bit-flip rate = 0.1												
MSTR	1.25	1.53	1.38	2.42	4.95	4.70	4.82	11.55	0.92	1.76	1.21	2.67
DAAN	0.84	1.59	1.10	2.26	3.62	3.60	3.61	13.15	1.17	1.39	1.27	2.78
QARM	4.64	3.33	3.88	3.74	10.28	8.80	9.48	13.17	3.01	3.33	3.16	3.56
Missing rate = 0.4, AWGN intensity = 0.8, Bit-flip rate = 0.1												
MSTR	1.44	1.01	1.19	2.32	4.25	6.88	5.26	11.72	0.87	1.43	1.08	2.48
DAAN	0.69	1.48	0.94	2.12	3.46	4.79	4.02	11.90	1.03	1.37	1.18	2.74
QARM	3.87	3.10	3.45	3.42	9.67	8.14	8.84	11.67	2.90	2.66	2.77	2.83
Missing rate = 0.4, AWGN intensity = 0.9, Bit-flip rate = 0.1												
MSTR	0.99	1.28	1.12	2.12	3.86	6.70	4.90	11.52	1.10	1.19	1.14	2.45
DAAN	0.82	1.04	0.92	1.94	3.24	4.93	3.91	12.05	0.92	1.25	1.06	2.62
QARM	3.08	2.46	2.74	3.13	8.31	8.26	8.28	11.70	2.20	2.85	2.48	3.08